

DRAW: Deep Robustness-Adjustable Watermarking

Han Fang^{ID}, Kejiang Chen^{ID}, Yupeng Qiu, Jiyi Zhang^{ID}, Weiming Zhang^{ID}, and Ee-Chien Chang^{ID}

Abstract—Robustness stands as a vital attribute of watermarking schemes. To ensure comprehensive robustness across different scenarios, deep learning-based watermarking frameworks often integrate a combined noise layer during the training phase. This layer encompasses various distortions, empowering the model to adeptly withstand these distortions, thereby guaranteeing broad-spectrum robustness. However, a notable limitation of the current approach lies in its inflexibility of robustness adjustment. Once trained, all forms of robustness become intertwined, hindering independent adjustments to each kind of robustness. This lack of flexibility significantly curtails the adaptability of the watermarking system. To address such limitations, this article proposed DRAW, a two-stage deep multi-robust-adjustable watermarking framework, which can effectively decouple each robustness for individual adjusting. In stage-1, we randomly generate several orthogonal masks to guide the embedding regions of each robust feature. Based on the masks, we parallel-train several watermark networks for every single distortion. In stage-2, we fix the well-trained feature embedding network in stage-1 and train a transformer-based decoder to adapt to different watermark embedding features. In this way, each robustness can be adjusted independently by weighting the embedding strength of different regions, and all the different features can be extracted based on the powerful decoder. Besides, we also provide a frequency loss to ensure a better “transparency-robustness” trade-off. Experiments indicate the superior performance in robustness as well as the adjustability compared with the state-of-the-art framework and the strong generalizability of the proposed framework.

Index Terms—Deep learning-based watermarking, robustness, adjustability.

I. INTRODUCTION

WATERMARKING technique, as an important means to realize copyright protection and leak source tracing, has been widely studied [1], [2], [3], [4]. The most important property of the watermarking scheme is robustness, ensuring the lossless recovery of watermarks even amidst image distortions. Traditional watermarking schemes often embed the watermark into transform domain coefficients, since the transform domain coefficients are more stable to different distortions. Commonly used domains include discrete cosine transform (DCT) domain [5], [6], discrete wavelet transform domain (DWT) [7] and discrete Fourier transform domain (DFT) [8], [9].

Received 7 February 2026; revised 16 June 2026; accepted 8 July 2026. Date of publication 17 July 2026; date of current version 1 September 2026. This work was supported in part by the National Natural Science Foundation of China under Grant U2336206, Grant 62472398, Grant 62121002, and Grant 62402469. (Corresponding authors: Kejiang Chen; Ee-Chien Chang.)

Han Fang, Kejiang Chen, and Weiming Zhang are with the University of Science and Technology of China, Hefei, Anhui 230052, China (e-mail: fanghan@ustc.edu.cn; chenkj@ustc.edu.cn; zhangwm@ustc.edu.cn).

Yupeng Qiu, Jiyi Zhang, and Ee-Chien Chang are with the Department of School of Computing, National University of Singapore, Singapore 119077 (e-mail: changec@comp.nus.edu.sg).

Digital Object Identifier 10.1109/TDSC.2026.3714391

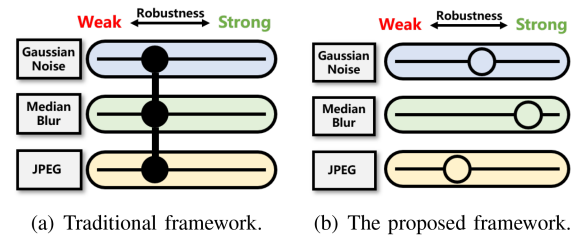


Fig. 1. The robustness adjustment comparison after training with combined noise layer (“JPEG compression”+“Gaussian Noise”+“Median Blur”).

For deep-learning-based watermarking framework [10], [11], the typical structure includes an encoder that embeds the watermark into the host image, a noise layer that distorts the watermarked image, and a decoder that aims to extract the watermark from the distorted image. The robustness is obtained by introducing different distortions in the noise layer and end-to-end training of these three components. The characterization of the learned robust feature to be embedded into the host image (i.e., the residual signal that contains the watermark) is highly related to the noise layer design. After training, the robustness level can be adjusted by varying the embedding strengths of these features [12], [13].

Generally, to achieve broad-spectrum robustness for various scenarios, a combined noise layer that contains multiple distortions is often adopted in the training stage. Through iterative training with each distortion in the noise layer, the network gradually learns a common feature robust to all distortions. However, a potential limitation of combined noise training is that once the model is trained, all types of robustness become coupled to a single robust feature, such that adjusting the embedding strength will inevitably affect all types of robustness simultaneously. For example, in a watermarking system with JPEG robustness, Gaussian noise robustness, and median filtering robustness, weakening the robustness of Gaussian noise will also weaken the robustness of JPEG compression, as illustrated in Fig. 1, a coupling effect that occurs at test time due to unified embedding strength.

Enabling the independent adjustability of each individual robustness factor is crucial in real-world applications, as it offers users greater flexibility in meeting diverse demands, particularly when confronted with similar but slightly different distortion channels. One typical scenario is image sharing across various online social network (OSN) platforms. To protect copyright, users often watermark images before uploading them. However, different OSN platforms may apply similar yet distinct image processing operations to uploaded images. In this case, the watermark robustness of the uploaded images shall be adjusted

accordingly. For instance, as outlined in [14] and corroborated by our experiments, platforms like Facebook, WeChat, and WhatsApp employ different image enhancement processes (e.g., filtering) and varying levels of JPEG compression (QF=65/80/71) on uploaded images. Consequently, it becomes imperative to tailor the JPEG compression robustness and image filtering robustness of watermarked images for each platform. Detailed experiments illustrating this scenario are presented in Section IV-F.

Therefore, in this paper, we look into the adjustability aspect: for a well-trained watermarking system with multiple-distortion robustness, can we realize the independent adjustability of each specified robustness? Techniques derived from this inquiry could significantly broaden the scope of the application and offer customizable options tailored to different scenarios.

Essential Requirements. Two essential requirements shall be addressed in realizing the adjustability.

- 1). For specific distortions, the corresponding robustness shall be adjusted in a simple way after training.
- 2). For multiple distortions, the robustness features of each two distortions shall be orthogonal.

Proposed Solution. For 1), [12] and [13] have demonstrated that the specified robustness can be adjusted by weighting the embedding strength of a well-trained robust feature. Therefore, we aim to explore better features of specified distortion for further adjustment. Motivated by the traditional watermarking schemes, we proposed a frequency loss, which encourages the watermark feature to be embedded into intermediate frequency (IF) coefficients, striking a better balance between visual quality and robustness. For 2) we proposed to utilize orthogonal mask-guided embedding to satisfy the feature orthogonality. We pre-define several orthogonal masks to guide the embedding, and each mask region is embedded with one kind of robust feature. In this way, every kind of robustness can be adjusted by weighting the embedding strength of the corresponding mask region.

Building on the analysis, we proposed DRAW, a two-stage **deep robustness adjustable watermarking** framework. Specifically, in stage-1, we pre-define several orthogonal masks and train multiple watermarking networks in parallel. Each network is trained with a specific mask and only handles one type of distortion. This approach ensures that different robust features are embedded into different masked regions, allowing for independent adjustment of each type of robustness. The final watermarked image is generated by embedding all the mask-guided robust features together.

On the extraction side, since different distortions lead to different robust features, we require a powerful decoder that can adapt to all these features and accurately extract the watermark. To achieve this, in stage-2, we design a transformer-based architecture consisting of a transformer-based feature extractor and multiple well-trained decoders from stage-1. Each well-trained decoder is specialized in handling a specific type of distortion. The transformer-based feature extractor provides these decoders with different attention based on the distortion of the image, enabling the extraction process to be conducted in the most suitable mechanism. In this stage, we embed the watermark using the well-trained encoders and corresponding masks from stage-1, then only fine-tune the transformer-based decoder with various

distortions. Thanks to the attention mechanism, the powerful decoder can effectively adapt to all the distorted features and accurately extract the watermark.

Contributions.

- 1) We point out a new property of the robust watermarking framework, that is, the adjustability of multi-distortion robustness. Techniques derived could effectively expand the application scope of the current watermarking framework and provide diversified options for different scenarios.
- 2) We propose a two-stage framework to achieve adjustability. Stage-1 effectively trains several encoders with orthogonal masks for different distortions and stage-2 trains a powerful transformer-based decoder to extract these features. Besides, we also propose a bandpass-based frequency loss to better trade off the visual quality and robustness.
- 3) Extensive experiments indicate the strong robustness of the proposed scheme and the superior adjustability compared with the state-of-the-art DNN-based watermarking schemes.

II. RELATED WORK

A. Traditional Watermarking

Traditional watermarking schemes often embed the watermark into the transform domain. Commonly used transform domain includes DCT domain, DWT domain and DFT domain. Suhail et al. [15] proposed a DCT domain-based watermarking framework to resist JPEG compression distortion. By modifying the intermediate frequency (IF) coefficients using quantization index modulation according to the quantization table, the watermark signal is well reserved after JPEG compression. Daren et al. [7] proposed a DWT domain-based watermarking framework that embeds the watermark into LH sub-bands. Kang et al. [16] proposed embedding the watermark into DFT domain coefficients to resist the print-scanning distortions. Since the IF coefficients can better trade off the visual quality and the robustness, embedding the watermark into such coefficients is a common sense operation in traditional watermarking techniques. In this paper, we also draw on this insight and propose an IF bandpass-based frequency domain loss to guarantee a better trade-off in deep-learning-based frameworks.

B. Deep-Learning-Based Watermarking

The typical Encoder-Noise layer-Decoder-based (END-based) architecture was first proposed by [11], where the robustness is obtained by training with the noise layer. By setting different distorting operations in the noise layer, the robustness against different types of distortions can be effectively guaranteed. For example, Jia et al. [12] proposed a mini-batch strategy with a mixed noise layer, which contains the real JPEG and simulated JPEG to achieve JPEG compression robustness. Tancik et al. [17] proposed a noise layer for print-shooting robustness by simulating the distortion with several distortions. Similarly, Fang et al. [18] proposed a noise layer for screen-shooting channel with simulated perspective distortion,

illumination distortion, and moiré distortion. Wengrowski et al. [19] proposed to utilize a simulated network as the noise layer to guarantee the screen-shooting robustness. To train such a network, they generate million-level data pairs. In addition to utilizing certain operations as the noise layer, Luo et al. [20] proposed to use an adversarial network as the noise layer to generate the worst distorted image for better robustness. Ma et al. [13] proposed a flow-based architecture that combines invertible and non-invertible mechanisms.

C. Visual Transformer

Transformer was first proposed in 2017 for natural language processing (NLP) tasks [21]. Later on, it has been applied to different computer vision tasks, such as object detection [22], [23], classification [24], re-identification [25]. The core module of the Transformer is the attention mechanism, which includes self-attention, mask attention and cross-attention. Attention is designed to capture the internal relevance of different features, giving necessary weight to different information. It greatly solves the long-distance dependence problem by calculating the mutual influence between different patches of the image [26]. In this paper, we designed a transformer-based decoder to obtain different attention with different masks for better performance.

III. PROPOSED FRAMEWORK

A. Challenges and Motivations

As introduced before, this paper aims to achieve a multi-robustness adjustable watermarking framework. Since single distortion robustness can be adjusted by weighting the embedding strength of the robust feature, the challenges of multi-robustness adjustability rely on two main aspects: **A)** In the embedding part, the orthogonality of different robust features should be guaranteed such that we can separately adjust each robustness. **B)** Since different distortions will result in different robust features, in the extraction part, a powerful decoder should be designed and trained to adapt to all these features.

Challenge **A)** can be addressed by mask-guided embedding with pre-designed orthogonal masks. Before training, we generate a batch of mutually orthogonal masks in advance. For each distortion, we choose one mask to guide the embedding regions such that the targeted robustness can be adjusted by only weighting the features in the corresponding mask. Since we have already trained coupled encoders and decoders for each distortion, for challenge **B)**, we can make full use of the well-trained decoders to extract these distorted features. The only bottleneck is how to automatically integrate the features of these decoders based on different distorted images. Encouragingly, such a bottleneck can be tackled by the transformer. Since after specific distortion, only the corresponding robust features can be preserved, and such features maintain similar components, we can effectively use the transformer to analyze the distorted image to generate attention for each decoder. Based on the attention mechanism, the watermark in different distorted images can be adaptively and automatically extracted.

B. Overview

The proposed framework mainly contains two training stages, as shown in Fig. 2. Before stage-1 training, we first design n mutually orthogonal masks $\{\mathcal{M}_1, \mathcal{M}_2, \dots, \mathcal{M}_n\} \in \mathbb{M}$. Then we will parallel train n encoders $\{\mathcal{E}_1, \mathcal{E}_2, \dots, \mathcal{E}_n\} \in \mathbb{E}$ and decoders $\{\mathcal{D}_1, \mathcal{D}_2, \dots, \mathcal{D}_n\} \in \mathbb{D}$ with n different noise layers $\{\mathcal{N}_1, \mathcal{N}_2, \dots, \mathcal{N}_n\} \in \mathbb{N}$. During training, each mask $\mathcal{M}_i, i \in [1, n]$ will guide an embedding region of an corresponding encoder $\mathcal{E}_i, i \in [1, n]$. After stage-1 training, we can get n well-trained encoders. Then in stage-2, we fixed these encoders $\mathcal{E}_i^\Delta, i \in [1, n]$ and utilize all of them along with $\mathbb{M}$ to generate the watermarked image to train a transformer-based decoder $\mathcal{D}_T$. The noise layer used in stage-2 is randomly selected from $\mathbb{N}$ for each iteration. In the testing frame, the embedding process is conducted with $\mathbb{E}$ and masks $\mathbb{M}$, and each encoder $\mathcal{E}_i$ will correspond to a strength factor α_i to adjust the embedding strength. After distortion, the distorted image is fed to $\mathcal{D}_T$ to adaptively extract the watermark. In high level sense, stage-1 is responsible for learning diverse and disentangled robustness representations; Stage-2 is responsible for adaptively integrating and utilizing these learned robustness features during decoding. In this way, the transformer-based decoder acts as a feature routing and fusion module, enabling adaptive decoding under different distortion conditions.

C. Components

1) *Orthogonal Mask*: Given a host image $I_H \in \mathbb{R}^{W \times H \times 3}$, we generate n binary spatial masks $\{M_1, M_2, \dots, M_n\}$, where each $M_i \in \{0, 1\}^{W \times H}$ has the same spatial size as the image. Each mask corresponds to one specific robustness feature (e.g., JPEG, blur, noise). To ensure that different robustness features are embedded into non-overlapping regions, the masks are designed to be spatially orthogonal and mutually exclusive:

$$M_i \odot M_j = 0, \quad \forall i \neq j, \quad (1)$$

where $\odot$ denotes the Hadamard (element-wise) product and the result is a zero matrix, indicating no overlap between any two masks.

Additionally, the masks collectively cover the entire image:

$$\sum_{i=1}^n M_i(x, y) = 1, \quad \forall (x, y), \quad (2)$$

ensuring that every pixel belongs to exactly one robustness sub-region. For example, when $n = 4$, each M_i has approximately 1/4 of its entries set to 1, forming a non-overlapping and complete partition of the image.

2) *Encoder*: Every specific encoder $\mathcal{E}_i$ only consists of a full-connection layer, which receives the watermark message $m \in \mathbb{R}^l$ as input and output a feature with length $W \times H \times 3$. Then the feature will be reshaped to the same size as the host image and further masked by $\mathcal{M}_i$ to generate the robust feature $\mathcal{R}_i \in \mathbb{R}^{W \times H \times 3}$. The watermarked image $I_i^{em} \in \mathbb{R}^{W \times H \times 3}$ is the sum of the host image I_H and the robust feature $\mathcal{R}_i$. The whole encoding process can be formulated by:

$$I_i^{em} = I_H + \mathcal{R}_i = I_H + \mathcal{E}_i(m) \otimes \mathcal{M}_i \quad (3)$$

where $\otimes$ indicates the Hadamard product operation.

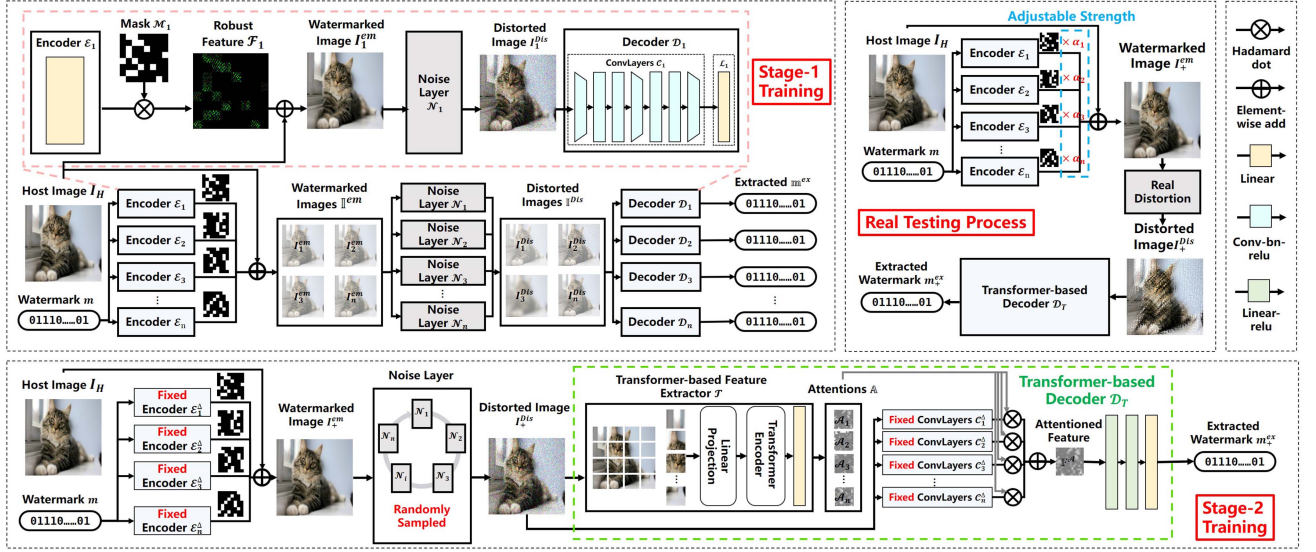


Fig. 2. The framework of DRAW. It consists of two main stages, the first stage trains several encoders and decoders with different robustness in parallel. During the training process, each embedded feature is guided by a mask and each two masks corresponded to different distortions that are orthogonal. In stage-2, the well-trained encoders in stage-1 are fixed to generate the whole embedded features, then a transformer-based decoder is trained to extract all the embedded features and losslessly recover the embedded watermark. In the real testing process, the paralleled encoders and the transformer-based decoder are served to embed the watermark and extract the watermark.

3) *Noise Layer*: There will be n noise layer $\mathcal{N}_i, i \in [1, n]$ in the whole training process, each noise layer contains one type of distortion. In stage-1 training, each noise layer $\mathcal{N}_i$ will targetedly distorted the watermarked image I_+^{em} to generate the distorted image $I_+^{Dis} \in \mathbb{R}^{W \times H \times 3}$, where $I_+^{Dis} = \mathcal{N}_i(I_+^{em})$. As for stage-2, the noise layer for each iteration is randomly sampled from $\mathbb{N}$.

4) *Decoder*: Each decoder $\mathcal{D}_i$ in stage-1 contains two main parts: the convlayers $\mathcal{C}_i$ and a full connection layer $\mathcal{L}_i$, where $\mathcal{C}_i$ is consisted of 7 “Conv-BN-ReLU” layers. After distorting with the noise layer, I_+^{Dis} is fed into the decoder $\mathcal{D}_i$ for watermark extraction. The output of $\mathcal{D}_i$ is the extracted watermark m_i^{ex} . The extraction process can be formulated by:

$$m_i^{ex} = \mathcal{D}_i(I_+^{Dis}) = \mathcal{D}_i(\mathcal{N}_i(I_+^{em})) \quad (4)$$

5) *Transformer-Based Decoder*: The purpose of stage-2 is to train the powerful transformer-based decoder $\mathcal{D}_T$. After stage-1, we can obtain n well-trained encoders. We fixed them in stage-2 denoted as $\mathcal{E}_i^A, i \in [1, n]$. Then we use all the $\mathcal{E}_i^A$ along with the corresponding $\mathcal{M}_i$ to generate the composited watermarked image I_+^{em} , where

$$I_+^{em} = I_h + \sum_{i=1}^n \mathcal{E}_i(m) \otimes \mathcal{M}_i \quad (5)$$

Then I_+^{em} is distorted by a randomly sampled noise layer to generate the distorted image I_+^{Dis} . Then I_+^{Dis} is fed into $\mathcal{D}_T$ to extract the watermark m_+^{ex} .

$\mathcal{D}_T$ contains two main parts: a transformer-based feature extractor $\mathcal{T}$ and n fixed convlayers $\mathcal{C}_i^A$ that are well-trained in stage-1. For $\mathcal{T}$, we utilize the traditional architecture of ViT [21] to generate the latent feature of the distorted image. Specifically, the input image is first divided by several blocks with size $B \times B$ and each block will be linearly projected into a vector named

patch embedding. Then each embedding will be added with the position embedding and further fed into the transform encoder, which consists of alternating layers of multi-head self-attention and MLP blocks. The output of the transform encoder will contain the valuable dependency information of these blocks. Then a full-connection layer is applied to process the latent feature into n attention maps $\{\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_n\} \in \mathbb{A}$. Each attention map $\mathcal{A}_i$ maintains the same size with $\mathcal{C}_i^A(I_+^{Dis})$. $\mathcal{A}_i$ will weight the corresponding extracted feature of $\mathcal{C}_i^A$ and generate the attention features $\{\mathcal{F}_1^A, \mathcal{F}_2^A, \dots, \mathcal{F}_n^A\} \in \mathbb{F}^A$, where:

$$\mathcal{F}_i^A = \mathcal{C}_i^A(I_+^{Dis}) \otimes \mathcal{A}_i$$

$\mathbb{F}^A$ will be further processed by three full connection layers with two “ReLU” activate functions to generate the final extracted watermark m_+^{ex} . The specific structure is illustrated in supplementary materials.

D. Loss Function

The total loss function is composed of three main parts: the image loss $\mathcal{L}_I$, the message loss $\mathcal{L}_M$ and the frequency loss $\mathcal{L}_F$.

1) *Image Loss*: The image loss $\mathcal{L}_I$ aims to ensure the visual similarity of the watermarked image and the host image. To achieve that, the mean squared error (MSE) loss is first applied in stage-1 training, which can be formulated by:

$$\mathcal{L}_I(I_+^{em}, I_h) = \|I_+^{em} - I_h\|^2 \quad (6)$$

2) *Message Loss*: The message loss $\mathcal{L}_M$ is essential to ensure robustness, which guarantees the extracted message to be the same as the embedded message. $\mathcal{L}_M$ is defined as follows:

$$\mathcal{L}_M(m_i^{ex}, m) = \|m_i^{ex} - m\|^2 \quad (7)$$

3) *Frequency Loss*: According to [9], embedding the watermark in the IF coefficients can well balance the visual quality

and the robustness. Therefore, we proposed a frequency loss $\mathcal{L}_F$, which gives a bandpass constraint on the frequency of the embedded features. We define the frequency loss $\mathcal{L}_F$ as follows:

$$\mathcal{L}_F(I_i^{em}, I_h) = \frac{1}{WH} \sum_{u=0}^{W-1} \sum_{v=0}^{H-1} \mathcal{B}_P |F_i^{em}(u, v) - F_H(u, v)|^2 \quad (8)$$

where $\mathcal{B}_P$ is a bandpass mask, F_i^{em} and F_H indicate the Fourier frequency coefficients corresponding to the image I_i^{em} and I_h . $\mathcal{B}_P$ is with the same size as I_h and can be defined as:

$$\mathcal{B}_P(x, y) = \begin{cases} \tau, & \text{if } th_l \leq \sqrt{(x - \frac{W}{2})^2 + (y - \frac{H}{2})^2} \leq th_h \\ 1, & \text{else} \end{cases} \quad (9)$$

where τ is the weight within the bandpass. th_l and th_h are the lower bound and higher bound of the bandpass mask. The Fourier frequency coefficients F_i^{em} can be formulated by:

$$F_i^{em}(u, v) = \sum_{x=0}^{W-1} \sum_{y=0}^{H-1} I_i^{em}(x, y) \cdot e^{-i2\pi(\frac{ux}{W} + \frac{vy}{H})} \quad (10)$$

By setting appropriate th_l and th_h , the robust feature can be embedded in the IF coefficients.

4) *Total Loss*: For stage-1, the training loss is a weight sum of $\mathcal{L}_I$, $\mathcal{L}_M$ and $\mathcal{L}_F$:

$$\mathcal{L}_{Total} = \lambda_1 \mathcal{L}_I(I_i^{em}, I_h) + \lambda_2 \mathcal{L}_M(m_i^{ex}, m) + \lambda_3 \mathcal{L}_F(I_i^{em}, I_h) \quad (11)$$

where λ_1 , λ_2 and λ_3 are weights to balance the losses. As for stage-2, only the message loss $\mathcal{L}_M(m_i^{ex}, m)$ is applied to train $\mathcal{D}_T$.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Experimental Settings

1) *Dataset and Settings*: In this paper, we utilize MS COCO [27] as the training dataset. As for testing, we choose the classical USC-SIPI image dataset [28]. For all the images used for training and testing, the size is set as $128 \times 128 \times 3$. The watermark message is randomly sampled from the binary distribution with length $l = 30$. The values of parameters λ_1 and λ_2 are set as 10 and 1 in the first 10 epochs and changed to 1 and 10 in the rest iteration. For λ_3 , we fixed it as 0.001 in all the training processes. The value of τ we choose is 100, and th_l and th_h are set as 20 and 60 for all the robustness training. The framework is implemented by PyTorch [29] and is run on one Tesla V100. For parameter optimization, we utilize Adam [30] with default hyperparameters.

2) *Baseline and Benchmark*: We compare the performance of the proposed method with several state-of-the-art (SOTA) watermarking methods: HiDDeN [11], DADW [20], MBRS [12], CIN [13], PIMoG [18], StegaStamp [17], FIN [31], InvisMark [32] and LWMark [33]. We train the proposed framework as well as the compared methods (HiDDeN, MBRS, CIN, FIN, InvisMark and LWMark) with four noise layers (“Gaussian Noise”, “Median Blur”, “JPEG Compression” and “Geometric Distortion”). These distortions cover compression, noise, filtering, and spatial transformation, which are the primary sources of

degradation in practical image transmission and editing scenarios. Therefore, they serve as a representative set of orthogonal robustness dimensions, making them suitable for evaluating the proposed robustness-adjustable framework.

For DADW, PIMoG and StegaStamp, since they are not designed for combined noise, we follow the training setting or their paper and directly utilize the pre-trained model^{1,2}. For the proposed framework, we randomly generate 4 block-wise orthogonal masks with block size 16. Then we conduct the stage-1 and stage-2 training with these masks and test the performance on the final well-trained network. The final embedded image in the testing frame is a weighted form of (5) by 4 strength factor $\alpha_1, \alpha_2, \alpha_3$ and α_4 :

$$I_+^{em} = I_h + \sum_{i=1}^4 \mathcal{E}_i(m) \otimes \mathcal{M}_i \times \alpha_i$$

For the compared methods, we directly use the combined noise layer which contains these four distortions for training. In testing frame, the robustness of 8 types of distortions, including 4 types of training distortion (“Gaussian Noise”, “Median Blur”, “JPEG Compression” and “Geometric Distortion”) and 4 types of untrained distortion (“Cropout”, “Salt&Pepper Noise”, “Gaussian Blur” and “Color manipulation”) are evaluated, as shown in Fig. 4. “Geometric Distortion” contains the perspective distortion and the rotate distortion. “Color manipulation” includes contrast changes and brightness changes.

3) *Evaluations*: Three aspects of performance are evaluated in this paper: The visual quality of the watermarked images, the robustness of each distortion, and the adjustability of the multi-robustness. The visual quality is measured by the peak signal-to-noise ratio (PSNR), structural similarity index measure (SSIM) (a larger value represents better visual quality) and the LPIPS [34], which is a learned perceptual similarity metric, lower is better. For robustness, we directly utilize the extraction accuracy as the evaluation metric, the larger value indicates the higher robustness. We test the adjustability by adjusting the embedding strength of one kind of robustness and testing the robustness of other distortions. If changing one robustness heavily affects another robustness, the adjustability is poor.

B. Visual Quality

To subjectively evaluate the visual quality of the watermarked image, we give one example for each method, as shown in Fig. 3. Besides, we also provide the objective indicators, as shown in Table I.

In Fig. 3, different columns indicate different methods. The first row, the second row and the third row represent the host images, the embedded images and residuals respectively. Combining the visual results with Table I, we can see that the proposed method achieves consistently high visual quality in terms of PSNR, SSIM, and LPIPS, indicating that the embedded watermark introduces only minor perceptual distortion. The

¹<https://github.com/tancik/StegaStamp>

²<https://github.com/FangHanNUS/PIMoG-An-Effective-Screen-shooting-Noise-Layer-Simulation-for-Deep-Learning-Based-Watermarking-Netw>

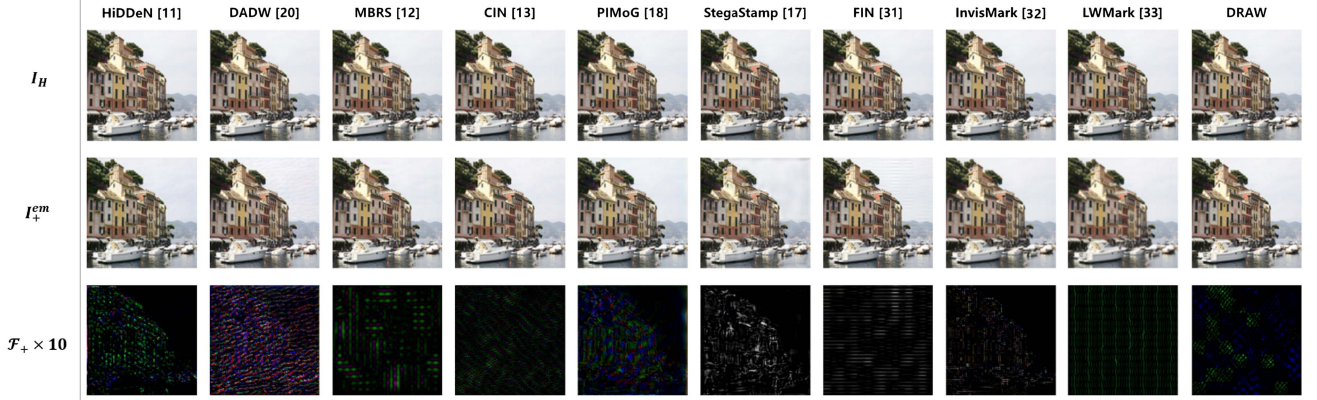


Fig. 3. The visual results of the watermarked image with different methods.

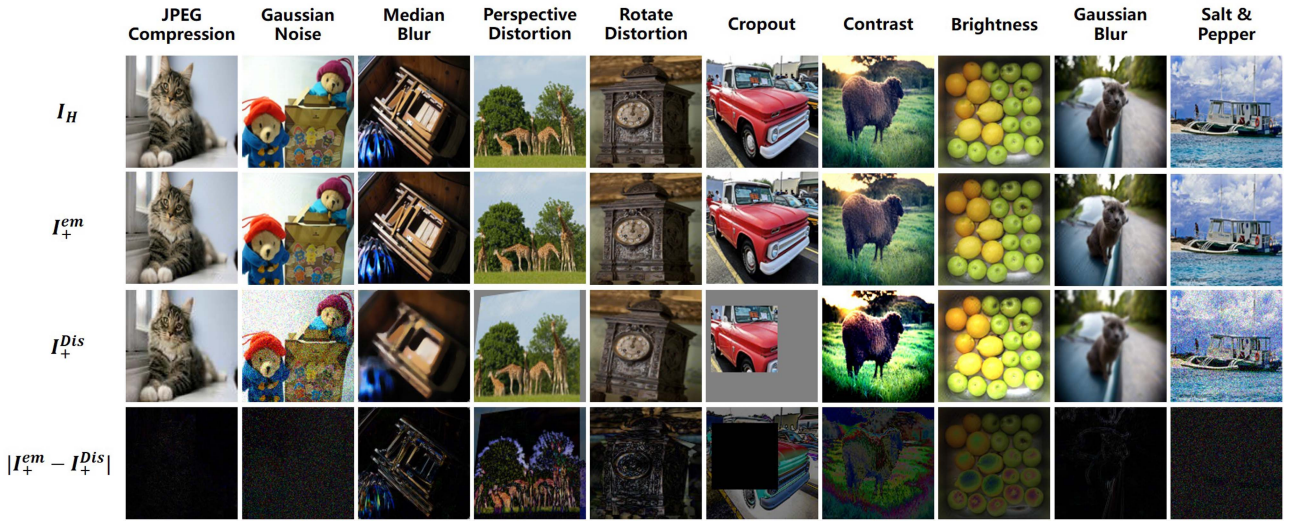


Fig. 4. The visual comparisons of different distortions. Each column indicates a type of noise. Different rows correspond to different images. Top: host images I_H s. Second row: watermarked images I_+^{em} s. Third row: distorted images I_+^{Dis} . Bottom: the difference $|I_+^{em} - I_+^{Dis}|$.

TABLE I
VISUAL QUALITY OF DIFFERENT METHODS

Methods	HiDDeN [11]	DADW [20]	MBRS [12]	CIN [13]	PIMoG [6]
PSNR(dB) $\uparrow$	31.23	32.05	40.19	40.00	36.28
SSIM $\uparrow$	0.9411	0.8823	0.9820	0.9673	0.9859
LPIPS $\downarrow$	0.0975	0.1657	0.0489	0.0466	0.0829
Methods	StegaStamp [17]	FIN [31]	InvisMark [32]	LWMark [33]	DRAW
PSNR(dB) $\uparrow$	39.86	38.63	40.24	39.17	40.37
SSIM $\uparrow$	0.9876	0.9817	0.9485	0.9641	0.9886
LPIPS $\downarrow$	0.0522	0.0707	0.0457	0.0505	0.0453

residual maps shown in Fig. 3 are magnified by a factor of $\times 10$ for visualization purposes, from which we can further observe that the watermark-induced differences remain visually subtle.

C. Robustness Test

The robustness test is performed on 8 distortions including 4 types of trained distortion (“Gaussian Noise”, “Median Blur”, “JPEG Compression” and “Geometric Distortion”) and 4 types of untrained distortion (“Cropout”, “Salt&Pepper Noise”,

“Gaussian Blur”, and “Color manipulation”). “Geometric Distortion” contains two types that are perspective distortion and rotate distortion. For “Color Manipulation”, we include the brightness and contrast changes in testing. All the tests are conducted under the same level of visual quality. The experimental results are shown in Table II.

As seen in Table II, the performance of the proposed method is significantly better than HiDDeN and DADW, where the average extraction accuracy is 20% higher than the compared method. For PIMoG and StegaStamp, the average extraction accuracy is 8% lower than DRAW. For MBRS, CIN, FIN and InvisMark, the average extraction accuracy is slightly lower than the proposed method. Such results indicate the strong robustness of the proposed method. Besides, we can see that for MBRS, CIN and InvisMark which are trained with a combined noise layer, although the extraction accuracy with Gaussian noise, median blur, and perspective distortion is maintained at a high level ($\geq 96.5\%$), the performance of JPEG compression robustness is not good. This reflects a potential limitation of the combined noise layer that it may not be easy to train a single feature that can be robust to all the distortions in the combined noise layer

TABLE II
ROBUSTNESS RESULTS OF DIFFERENT METHODS

Distortion	JPEG Compression QF = 50	Gaussian Noise $\sigma^2 = 0.05$	Median Blur w = 7×7	Perspective Distortion d = 4	Rotate a = 4°	Cropout r = 36%	Contrast s = 0.4	Brightness s = 0.4	Gaussian Blur $\sigma^2 = 0.05$	Salt& Pepper r = 0.05	Average
HiDDeN [11]	59.01%	57.32%	63.13%	86.94%	87.96%	88.55%	89.81%	90.02%	90.43%	56.51%	76.97%
DADW [20]	52.73%	50.41%	50.20%	67.51%	69.16%	80.20%	82.92%	82.31%	85.20%	50.20%	67.08%
MBRS [12]	75.42%	99.38%	97.92%	98.96%	99.79%	90.83%	99.61%	99.78%	99.79%	98.12%	95.96%
CIN [13]	81.88%	99.62%	99.84%	96.53%	99.62%	100%	100%	100%	100%	98.88%	97.64%
PIMoG [18]	85.24%	72.22%	96.98%	98.41%	99.39%	92.06%	99.68%	99.68%	96.83%	65.87%	90.64%
StegaStamp [17]	91.25%	87.81%	90.23%	92.08%	88.21%	84.29%	91.02%	90.31%	89.50%	88.52%	89.32%
FIN [31]	98.59%	99.03%	90.55%	86.53%	94.64%	69.12%	80.21%	99.18%	92.71%	93.68%	90.41%
InvisMark [32]	83.67%	86.85%	99.89%	100.00%	99.98%	82.15%	99.97%	99.90%	99.98%	91.70%	94.50%
LWMark [33]	99.61%	99.60%	100.00%	53.52%	54.20%	63.57%	100.00%	100.00%	100.00%	99.90%	87.04%
DRAW	96.25%	99.13%	97.71%	97.67%	99.37%	96.87%	99.79%	99.79%	99.80%	96.46%	98.28%

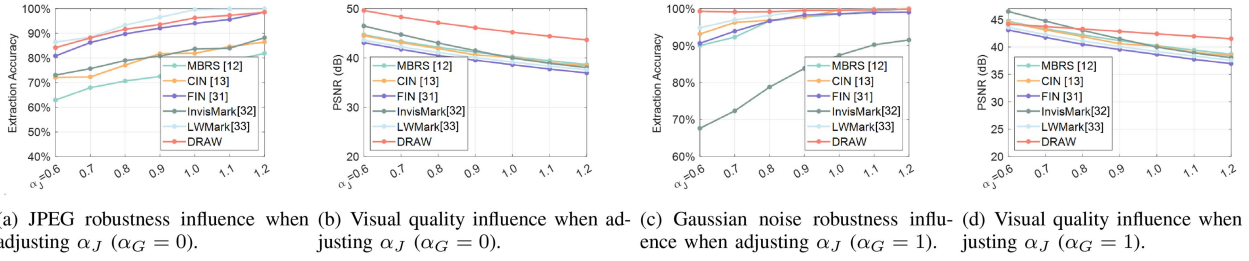


Fig. 5. The influence of adjusting α_J regarding the “Gaussian Noise” and “JPEG compression” distortions.

especially when the distortion number increases. Similar results can be found in FIN and LWMark, although the JPGE robustness is achieving 99%, the cropout robustness is quite low. But for the proposed method, as we see in Table II, the robustness of the trained distortions all stay at a high level, where the extraction accuracy is all higher than 96.2%. We attribute the reason to the orthogonal feature embedding which can effectively ensure the various robustness. Besides, we also see that the robustness of untrained image processing distortions is also good, which indicates the great robustness transferability of the proposed method. We believe the transformer-based decoder is the key to realizing such performance since the transformer architecture can effectively give different attention maps, which provide better guidance to the pre-trained decoders for further extraction.

D. Adjustability Test

1) *Adjustability Comparison:* In this section, we mainly show the adjustability performance of the proposed methods. The evaluation includes two parts. The first part compares the proposed method with five model training with a combined noise layer: MBRS, CIN, FIN, InvisMark and LWMark. The combined noise layer contains four distortions “Gaussian Noise”, “Median Blur”, “JPEG Compression” and “Geometric Distortion”. The strength factor of each robustness is noted as: α_J for JPEG robustness, α_G for Gaussian noise robustness, α_M for median blur robustness and α_O for geometric distortion robustness. After training, we first set α_G , α_M and α_O to 0 and change the value of α_J to adjust the robustness of “JPEG

compression”. Then we test the PSNR value of the method and the extraction accuracy after JPEG compression (QF=50). Besides, we also set $\alpha_G = 1$ and show PSNR and the “Gaussian Noise” ($\sigma^2 = 0.05$) robustness changing with the adjustment α_J . Since the compared methods only have one strength factor α_C which controls all the robustness, so we adjust the “JPEG compression” robustness with α_C in these methods. The results are shown in Fig. 5(a)–(b) for “JPEG compression” robustness and Fig. 5(c)–(d) for “Gaussian Noise” robustness.

It can be seen that the proposed methods outperform the compared methods in adjustability. For “JPEG compression” robustness adjustability test, all the methods show the same trade in extraction accuracy and PSNR. As the increase of α_J , the PSNR decreases but the extraction accuracy increases. It can be seen that the PSNR and extraction accuracy of the proposed method achieves the higher value in all different α_J values as shown in Fig. 5(a)–(b), such a result indicates the superior performance regarding the “JPEG compression” robustness. Besides, for the robustness of “Gaussian Noise”, we set $\alpha_G = 1$ and use α_J for adjusting and testing the corresponding performance. We can see that the changing of α_J will not influence the robustness of Gaussian noise with the proposed method, and the PSNR value is higher than the compared methods in most cases. But for MBRS, CIN, FIN, InvisMark and LWMark, the changing of JPEG robustness will simultaneously affect the robustness of “Gaussian Noise”. This indicates the poor adjustability of the compared methods in single robustness. For the proposed method, we can effectively adjust single robustness targetedly without affecting other robustness.

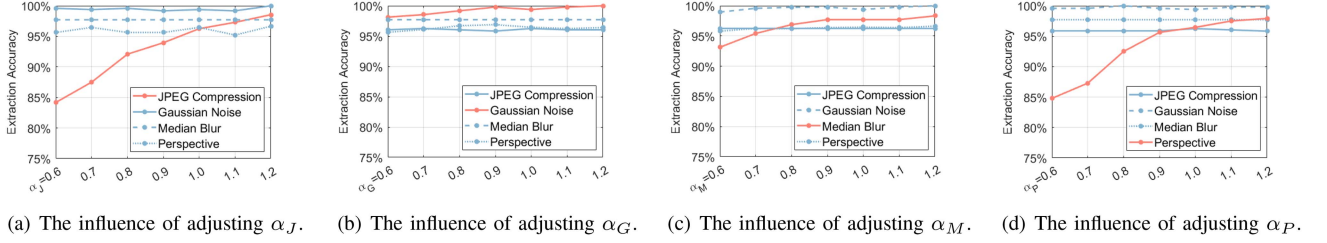


Fig. 6. The extraction accuracy results against different kinds of distortions under the adjustment of one kind of robustness.



Fig. 7. The visual results of black-box distortions.

2) *The Influence of Adjusting One Robustness on Other Robustnesses.*: Besides, we also test the influence of changing one single strength factor on other robustness, the results are shown in Fig. 6. It can be seen from Fig. 6(a) that the robustness of “Gaussian noise”, “median blur” and “perspective distortion” will not be affected by the changing of JPEG robustness, where the extraction accuracy with these three distortions remain stable in all α_J values. Similar results can be found in Fig. 6(b)–(d). These results indicate that the robustness of the proposed method can be separated and adjusted, the changing of one specific robustness will not influence other robustness.

E. Generalizability Test

To test the generalizability performance of the proposed method, we choose three types of black-box distortions: screen-shooting, instant messaging (IM) APP transmission and style transferring. The visual results of these distortions are shown in the supplementary materials. The screen-shooting distortion is conducted by “iPhone 12”, “Huawei P30 Pro” and “DELL P2319H”. The IM apps we choose are WeChat and WhatsApp. The style transferring we utilized is the crayon style and oil-painting style, which is done by Meitu.³ We visualize the black-box distortions in Fig. 7. The extraction results are shown in Table III.

As seen in Table III, for all the black-box distortions, the extraction accuracy achieves more than 95%, which indicates the great generalizability of the proposed methods.

³<https://pc.meitu.com/image>

TABLE III
THE EXTRACTION ACCURACY OF BLACK-BOX DISTORTIONS

Distortion	iPhone -DELL	Huawei -DELL	WeChat
Acc	95.83%	97.29%	98.00%
Distortion	WhatsApp	Crayon	Oil-Painting
Acc	100%	97.92%	98.54%

TABLE IV
EXTRACTION PERFORMANCE OF DIFFERENT APPS

α_J / α_M	1 / 0.1	1.5 / 0.1	1.5 / 0.5	1.5 / 1
PSNR(dB)	45.06	41.68	40.68	38.56
WeChat	90.00%	98.00%	100%	100%
Facebook	93.67%	96.33%	97.67%	100%
WhatsApp	96.67%	100%	100%	100%

F. Online Social Network Distortion Test

As described in the Introduction, a typical scenario of the robustness adjustability is image sharing in different online social network (OSN) APPs. The user may upload the watermarked image to different APPs, which often perform similar but different processing on the uploaded image, in this case, to protect the copyright of the uploaded images, the robustness shall be adjusted accordingly. For example, according to paper [14] and our experiments, the OSN APPs often conduct JPEG compression and filtering on the uploaded image, but the detailed procedure will be different in different APPs. Here we perform the experiments on 3 commonly used OSN APPs: WeChat, Facebook and WhatsApp. Then we use 10 images to conduct the robustness adjustability test. The watermark model we choose maintains the JPEG compression robustness with strength factor α_J and median filtering robustness with strength factor α_M . We embed the watermark with different α_J and α_M and upload the watermarked images into different APPs. Subsequently, we download the images and extract the watermark. The extraction accuracy is shown in Table IV. Besides, we also provide one example to show the difference in the distortion introduced by different APPs.

It is clear to see in Fig. 8 that different APPs will result in different distortions. Besides, from Table IV we can see that by adjusting the strength factor, we can effectively achieve the robustness against different OSN APPs’ distortion. When

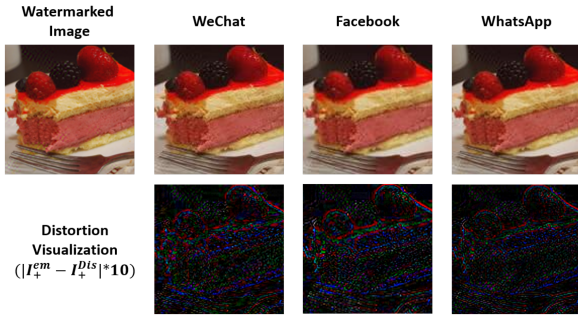


Fig. 8. The processing differences of different APPs.

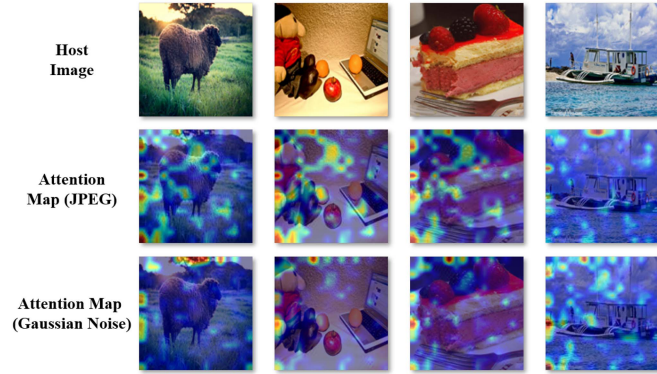


Fig. 9. Decoder attention maps under different distortion types. Distinct attention patterns suggest adaptive focus guided by distortion-specific features.

α_J / α_M is set as 1 and 0.1, the extraction accuracy after WeChat compression is only 90%, but for WhatsApp, it achieves 96.67%. Such a result indicates that WhatsApp produces the weakest distortions on the uploaded images, such a conclusion can also be observed in Fig. 8. Besides, to ensure the lossless extraction, we shall set different α_J and α_M for different APPs. For example, to ensure the WeChat robustness, α_J and α_M shall be set as 1.5 and 0.5, and for Facebook robustness, α_J and α_M shall be set as 1.5 and. In practice, we would give a mapping rule on the strength factor α and the robustness degree to guide the user adjustment. The more information about the channel distortion we know, the better adjustment can be made.

G. Time Cost of Transformer-Based Decoder

To evaluate the computational overhead introduced by our proposed transformer-based decoder, we conduct a runtime comparison between two decoding configurations: (1) Single-feature decoder: a baseline setting that uses a single encoder-decoder pair tailored to one distortion type (e.g., JPEG compression). (2) Transformer-based decoder: our proposed setting with parallel decoder branches and a transformer-based selection module that dynamically weights different robustness-specific features.

We report the average extraction time per image over 100 samples for both configurations.

As shown in Table V, the transformer-based decoding introduces only a marginal increase in runtime, from 1.4 ms to 6.5 ms per image, while significantly enhancing robustness adaptability.

TABLE V
AVERAGE EXTRACTION TIME PER IMAGE UNDER DIFFERENT DECODER SETTINGS

Decoder Type	Single Decoder	Transformer-based Decoder
Time (s/image)	0.0014	0.0065

This <10 ms delay remains well within acceptable limits for practical deployment scenarios, especially in offline provenance verification or batch-processing pipelines. These results confirm that our design achieves a favorable trade-off between performance and efficiency, enabling flexible multi-robustness decoding with minimal overhead.

H. Visualization of Distortion-Specific Decoding Attention

To gain deeper insights into the semantic distinctions learned by our distortion-specific encoders and transformer-based decoder, we perform a visual analysis of the decoder's attention behaviors under different distortion scenarios.

We focus on two representative distortions—JPEG compression and Gaussian noise—each paired with its corresponding encoder. For each case, we first embed the distortion-specific robustness feature into the host image, apply the respective distortion, and then extract the watermark using the proposed transformer-based decoder. During the decoding phase, we employ Grad-CAM [35] to obtain spatial attention heatmaps from the decoder's internal layers.

As shown in Fig. 9, the decoder exhibits clearly distinct attention patterns depending on the distortion type. In the JPEG case, the attention focuses more on smooth and low-frequency regions, whereas in the Gaussian noise case, it shifts toward more textured and edge-dense regions. This behavior suggests that the decoder has learned to assign different semantic decoding pathways in response to varying degradation types.

These findings validate our architectural design: the combination of distortion-specific encoders and a transformer-based selection decoder effectively promotes localized robustness representation learning, and enables adaptive routing during watermark extraction. This empirical evidence further strengthens the interpretability and motivation of our multi-branch robust watermarking framework.

I. Ablation Study

1) *The Importance of $\mathcal{L}_F$* : In this paper, we design a frequency loss for better trade-off the invisibility and robustness. In this section, we will show and discuss the importance of $\mathcal{L}_F$. We train the models with and without $\mathcal{L}_F$ and adjust the average robustness ($98 \pm 0.5\%$) in the same level, then we provide two visual examples to subjectively compare the visual results of different watermarked images, as shown in Fig. 10. It can be seen from the local enlarged image that without $\mathcal{L}_F$, the watermark embedding will create significant visual distortion (the obvious point-like distortion). But with $\mathcal{L}_F$, the embedding will concentrate on the IF coefficients, and the residual will be smoothly spread out, thus causing less visual distortions.

Fig. 10. The visual differences of with/without of $\mathcal{L}_F$.TABLE VI
VISUAL QUALITY OF DIFFERENT METHODS

Methods	PSNR (dB)	SSIM	LPIPS	NIQE
$w/o \mathcal{L}_F$	40.25	0.9888	0.0486	5.25
$w \mathcal{L}_F$	40.37	0.9886	0.0453	5.18

TABLE VII
EXTRACTION ACCURACY OF DIFFERENT DECODER ARCHITECTURES

Distortion	JPEG Compression QF = 50	Gaussian Noise $\sigma^2 = 0.05$	Median Blur $w = 7 \times 7$	Perspective Distortion $d = 4$	Salt & Pepper $r = 0.05$	Average
$\mathcal{D}_{w/oT}$	73.75%	82.71%	53.12%	84.17%	81.16%	74.87%
$\mathcal{D}_T$	96.25%	99.13%	97.71%	97.67%	96.46%	97.44%

Besides, to objectively show the effectiveness, we also provide the score of PSNR, SSIM, LPIPS and the natural image quality evaluator (NIQE), which calculates the no-reference image quality score for images, lower is better.

The results are shown in Table VI, it can be seen that under the same level of PSNR and SSIM, the LPIPS score and the NIQE score with $\mathcal{L}_F$ is lower than that without $\mathcal{L}_F$, which indicates that with $\mathcal{L}_F$, the watermarked images can maintain better visual quality. Such objective and subjective results indicate the importance and effectiveness of $\mathcal{L}_F$ in improving the transparency of the watermark.

2) *The Importance of the Transformer-Based Decoder:* In the extraction part, to cooperate with various embedding features, we design a transformer-based decoder $\mathcal{D}_T$, where the core is a transformer-based feature extractor $\mathcal{T}$. To show the importance of the transformer-based architecture, we conduct an experiment on only training one decoder for extractions. Specifically, after stage-1 training, we perform the same embedding operation in Stage-2. Then we utilized the decoder without transformer-based extractor to conduct the stage-2 training. The well-trained decoder is noted as $\mathcal{D}_{w/oT}$. Then we perform the extraction experiments. Since we fixed the embedding stage, the visual quality of watermarked images is totally the same in these two architectures. The extraction accuracy results are shown in Table VII.

The distortions we tested are JPEG compression with a quality factor 50, Gaussian noise with a variance 0.05, the median blur with window 7, perspective distortion with offset pixel 4 and salt & pepper noise with a ratio 0.05. It is clear to see that the performance of the decoder with the transformer-based extractor

TABLE VIII
GENERALIZATION PERFORMANCE OF THE PROPOSED ARCHITECTURE

Distortion	JPEG Compression QF = 50	Gaussian Noise $\sigma^2 = 0.05$	Median Blur $w = 7 \times 7$	Perspective Distortion $d = 4$	Salt & Pepper $r = 0.05$	Average
$MBRS$	75.42%	99.38%	97.92%	98.96%	90.83%	92.50%
$MBRS++$	95.65%	99.63%	98.12%	99.10%	92.15%	96.93%

is significantly better than that without the transformer-based extractor, where the extraction accuracy with $\mathcal{D}_{w/oT}$ is 20% lower than that with $\mathcal{D}_T$. Since different encoders will embed different features, one decoder cannot well adapted to all the features. However, with the transformer-based extractor, each “ConvLayers $\mathcal{C}_i^\Delta$ ” trained in stage-1 can be assigned with different attention, such that different features can be targetedly extracted by pre-trained decoders and further ensures the robustness for different distortions. Such results indicate the importance of the transformer-based architecture in ensuring various robustness.

3) *The Generalization of the Architecture:* In this paper, we proposed an architecture that contains orthogonal-mask-based encoders and a transformer-attention-based decoder, which greatly ensures overall robustness against various distortions. We believe that the architecture is general and can be applied to other schemes to improve their overall performance. To show the generalization of the architecture, we combined the architecture with MBRS [12] and conducted the experiments. Specifically, we replace the encoder and decoder structure in this paper with the structure proposed by MBRS [12], then conduct the same training process in this paper. The well-trained framework is denoted as $MBRS++$. Then we perform the robustness test on the $MBRS++$ and the original MBRS framework trained with combined noise layers. The results are shown in Table VIII.

It can be seen that for all the test distortion, the performance of $MBRS++$ is better than $MBRS$, where the average extraction accuracy of $MBRS++$ is 4% higher than $MBRS$. Such results greatly illustrate the generalizability of the proposed architecture in robustness improvements.

4) *Effect of Frequency Constraint Coefficient τ :* In the proposed bandpass-based frequency loss, we introduce a bandpass mask B_p to guide the watermark residual toward designated mid-frequency components. The mask B_p is parameterized by a coefficient τ , which controls the strength of the frequency-domain constraint. A larger τ enforces stronger concentration of watermark energy within the target frequency band.

To study the effect of τ , we conduct an ablation by varying $\tau = \{1, 10, 50, 100, 150, 200\}$.

Fig. 11 visualizes the results, where the first row shows the original image, the second row presents the watermarked image, the third row displays the watermark residual, and the fourth row illustrates the corresponding frequency spectrum of the residual.

As τ increases, the watermark residual becomes increasingly concentrated within the specified mid-frequency band, indicating stronger spectral regularization imposed by B_p . However, when τ exceeds 100, the frequency distribution of the residual saturates and exhibits no noticeable further change. Based on this observation, we set $\tau = 100$ in all experiments.

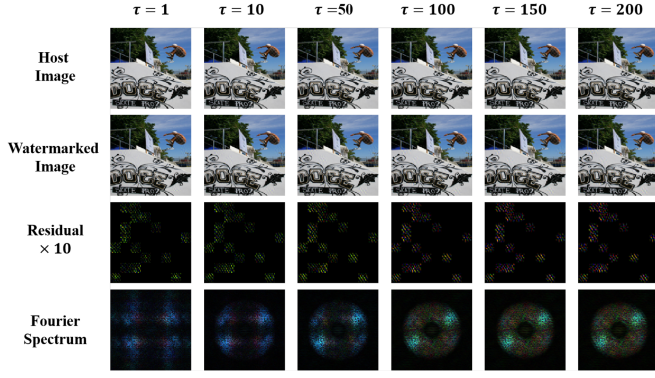
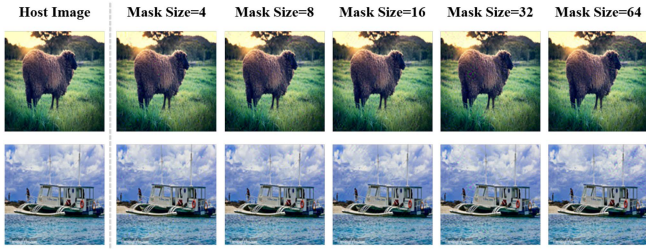
Fig. 11. The influence of τ .

Fig. 12. The visual quality of the watermarked images with different mask size.

TABLE IX
EXTRACTION ACCURACY WITH THE SETTING OF DIFFERENT MASK SIZE

Mask Size	PSNR	JPEG	GauNoise	MedFilter	GeoDistortion	Average
4	39.54 dB	90.00%	99.04%	91.43%	96.98%	94.36%
8	39.14 dB	95.24%	99.05%	93.49%	97.05%	96.21%
16	40.37 dB	96.25%	99.13%	97.71%	97.67%	97.69%
32	40.43 dB	95.22%	99.36%	98.89%	97.78%	97.81%
64	39.90 dB	95.56%	99.52%	97.94%	96.83%	97.46%

5) *Effect of Block Size in Mask Design*: In our framework, the binary mask determines the spatial scope of each robustness-specific embedding. To explore how the block size of this mask affects performance, we conducted controlled experiments using square masks of size 4×4 , 8×8 , 16×16 , 32×32 and 64×64 . During training, the distortion types were kept consistent (Gaussian Noise, Median Blur, JPEG Compression, and Geometric Distortion), while only the mask block size was varied. For each configuration, we measured both the visual quality (via PSNR and subjective visual results) and robustness (bit accuracy under the four distortions), results are shown in Fig. 12 and Table IX.

As we can see, the results reveal a clear trade-off: **Smaller blocks (e.g., 4, 8)** enable more dispersed and fine-grained embedding, leading to better imperceptibility. However, they offer less structural coherence, making the watermarked signal more fragile under block-based attacks such as JPEG and median filtering. **Larger blocks (e.g., 32, 64)** produce coarser embedding patterns, which enhance robustness but introduce visible block artifacts.

Among all choices, the 16×16 block achieves the best trade-off, offering visually imperceptible watermarking while maintaining strong robustness across all tested distortions. Based on this observation, we adopt block size 16 as the default configuration throughout the paper.

V. CONCLUSION AND FUTURE DIRECTION

This paper proposed an adjustable watermarking framework that can effectively realize the independent adjustment of different robustness. By training encoders with different orthogonal masks, the robust features for different distortions can be embedded into specific regions of an image. As a consequence, different robust features can be individually adjusted by weighting the embedding strength of different regions. On the extraction side, we proposed a transformer-based decoder to give attention maps for decoding according to the input distorted images. Based on the powerful feature extraction ability of the transformer structure, different robust features can be adaptively utilized for decoding. Besides, to guarantee a better trade-off between visual quality and robustness, we also provide a frequency loss to embed the watermark into the IF coefficients. Extensive experimental results show that our method can achieve strong robustness and generalizability, and significantly outperforms other SOTA methods in robustness adjustability.

One current limitation lies in the use of fixed, square-shaped masks for robustness partitioning, which are applied uniformly across all images regardless of content. While effective, such shape-agnostic masking may not fully leverage the spatial redundancy and semantic structure of each image. In future work, we plan to explore content-adaptive and image-specific mask generation strategies, aiming to further improve the trade-off between robustness and imperceptibility. Besides, looking forward, we believe that the automation of robustness strength selection is also a highly promising direction.

REFERENCES

- [1] M. Rathor, A. Anshul, and A. Sengupta, "Securing reusable IP cores using voice biometric based watermark," *IEEE Trans. Dependable Secure Comput.*, vol. 21, no. 4, pp. 2735–2749, Jul./Aug. 2024.
- [2] A. Anand and A. K. Singh, "Dual watermarking for security of COVID-19 patient record," *IEEE Trans. Dependable Secure Comput.*, vol. 20, no. 1, pp. 859–866, Jan./Feb. 2023.
- [3] M. He, H. Wang, F. Zhang, S. M. Abdullahi, and L. Yang, "Robust blind video watermarking against geometric deformations and online video sharing platform processing," *IEEE Trans. Dependable Secure Comput.*, vol. 20, no. 6, pp. 4702–4718, Nov./Dec. 2023.
- [4] Z. Qian, H. Zhou, X. Zhang, and W. Zhang, "Separable reversible data hiding in encrypted JPEG bitstreams," *IEEE Trans. Dependable Secure Comput.*, vol. 15, no. 6, pp. 1055–1067, Nov./Dec. 2018.
- [5] H.-J. Ko, C.-T. Huang, G. Horng, and W. Shiuh-Jeng, "Robust and blind image watermarking in DCT domain using inter-block coefficient correlation," *Inf. Sci.*, vol. 517, pp. 128–147, 2020.
- [6] H. Fang, W. Zhang, H. Zhou, H. Cui, and N. Yu, "Screen-shooting resilient watermarking," *IEEE Trans. Inf. Forensics Security*, vol. 14, no. 6, pp. 1403–1418, Jun. 2019.
- [7] H. Daren, L. Jiufen, H. Jiwu, and L. Hongmei, "A DWT-based image watermarking algorithm," in *Proc. IEEE Int. Conf. Multimedia Expo*, 2001, pp. 80–80.
- [8] M. Urvoy, D. Goudia, and F. Atrousseau, "Perceptual DFT watermarking with improved detection and robustness to geometrical distortions," *IEEE Trans. Inf. Forensics Security*, vol. 9, no. 7, pp. 1108–1119, Jul. 2014.

- [9] X. Kang, J. Huang, and W. Zeng, "Efficient general print-scanning resilient data hiding based on uniform log-polar mapping," *IEEE Trans. Inf. Forensics Security*, vol. 5, no. 1, pp. 1–12, Mar. 2010.
- [10] S.-M. Mun, S.-H. Nam, H.-U. Jang, D. Kim, and H.-K. Lee, "A robust blind watermarking using convolutional neural network," 2017, *arXiv:1704.03248*.
- [11] J. Zhu, R. Kaplan, J. Johnson, and L. Fei-Fei, "HiDDeN: Hiding data with deep networks," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 657–672.
- [12] Z. Jia, H. Fang, and W. Zhang, "MBRS: Enhancing robustness of DNN-based watermarking by mini-batch of real and simulated JPEG compression," in *Proc. 29th ACM Int. Conf. Multimedia*, 2021, pp. 41–49.
- [13] R. Ma et al., "Towards blind watermarking: Combining invertible and non-invertible mechanisms," in *Proc. 30th ACM Int. Conf. Multimedia*, 2022, pp. 1532–1542.
- [14] W. Sun, J. Zhou, L. Dong, J. Tian, and J. Liu, "Optimal pre-filtering for improving Facebook shared images," *IEEE Trans. Image Process.*, vol. 30, pp. 6292–6306, 2021.
- [15] M. A. Suhail and M. S. Obaidat, "Digital watermarking-based DCT and JPEG model," *IEEE Trans. Instrum. Meas.*, vol. 52, no. 5, pp. 1640–1647, Oct. 2003.
- [16] X. Kang, R. Yang, and J. Huang, "Geometric invariant audio watermarking based on an LCM feature," *IEEE Trans. Multimedia*, vol. 13, no. 2, pp. 181–190, Apr. 2011.
- [17] M. Tancik, B. Mildenhall, and R. Ng, "StegaStamp: Invisible hyperlinks in physical photographs," 2019, *arXiv:1904.05343*.
- [18] H. Fang, Z. Jia, Z. Ma, E.-C. Chang, and W. Zhang, "PiMoG: An effective screen-shooting noise-layer simulation for deep-learning-based watermarking network," in *Proc. 30th ACM Int. Conf. Multimedia*, 2022, pp. 2267–2275.
- [19] E. Wengrowski and K. Dana, "Light field messaging with deep photographic steganography," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 1515–1524.
- [20] X. Luo, R. Zhan, H. Chang, F. Yang, and P. Milanfar, "Distortion agnostic deep watermarking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 13548–13557.
- [21] A. Vaswani et al., "Attention is all you need," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2017, pp. 5998–6008.
- [22] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 213–229.
- [23] Y. Qian, M. Yang, X. Zhao, C. Wang, and B. Wang, "Oriented spatial transformer network for pedestrian detection using fish-eye camera," *IEEE Trans. Multimedia*, vol. 22, no. 2, pp. 421–431, Feb. 2020.
- [24] D. Dey, B. Chatterjee, S. Chakravorti, and S. Munshi, "Rough-granular approach for impulse fault classification of transformers using cross-wavelet transform," *IEEE Trans. Dielectr. Electr. Insul.*, vol. 15, no. 5, pp. 1297–1304, Oct. 2008.
- [25] M. Li, J. Liu, C. Zheng, X. Huang, and Z. Zhang, "Exploiting multi-view part-wise correlation via an efficient transformer for vehicle re-identification," *IEEE Trans. Multimedia*, vol. 25, pp. 919–929, 2023.
- [26] Y. Yang et al., "Transformers meet visual learning understanding: A comprehensive review," 2022, *arXiv:2203.12944*.
- [27] T.-Y. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.*, 2014, pp. 740–755.
- [28] U. Viterbi, "The USC-SIPI image database," 1977. Accessed: Mar. 2023. [Online]. Available: <http://sipi.usc.edu/database/>
- [29] R. Collobert, K. Kavukcuoglu, and C. Farabet, "Torch7: A Matlab-like environment for machine learning," in *Proc. BigLearn Neural Inf. Process. Syst. Workshop*, 2011, pp. 1–6.
- [30] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. 3rd Int. Conf. Learn. Representations*, 2015, pp. 1–15.
- [31] H. Fang, Y. Qiu, K. Chen, J. Zhang, W. Zhang, and E.-C. Chang, "Flow-based robust watermarking with invertible noise layer for black-box distortions," in *Proc. AAAI Conf. Artif. Intell.*, 2023, pp. 5054–5061.
- [32] R. Xu et al., "InvisMark: Invisible and robust watermarking for AI-generated image provenance," in *Proc. 2025 IEEE/CVF Winter Conf. Appl. Comput. Vis.*, 2025, pp. 909–918.
- [33] Y. Qiu, H. Fang, and E.-C. Chang, "Lightweight-mark: Rethinking deep learning-based watermarking," in *Proc. 42nd Int. Conf. Mach. Learn.*, 2025, Art. no. 2015.
- [34] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, "The unreasonable effectiveness of deep features as a perceptual metric," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 586–595.

- [35] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2017, pp. 618–626.



Han Fang received the BS degree from the Nanjing University of Aeronautics and Astronautics (NUAA), in 2016, and the PhD degree from the University of Science and Technology of China (USTC), in 2021. From 2021 to 2025, he is a research fellow with the National University of Singapore. Currently, he is a tenure-track professor with the University of Science and Technology of China. His research interests include image watermarking, information hiding, and adversarial machine learning.



Kejiang Chen received the BS degree from Shanghai University (SHU), in 2015, and the PhD degree from the University of Science and Technology of China (USTC), in 2020. Currently, he is an associate professor with the University of Science and Technology of China. His research interests include information hiding, image processing, and deep learning.



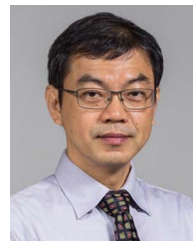
Yupeng Qiu received the BS degree from Sichuan University (SCU), in 2022. He is currently working toward the PhD degree with the National University of Singapore. His research interests include image watermarks, adversarial machine learning, and back-door attack.



Jiye Zhang received the bachelor's degree from the National University of Singapore, in 2018, and the PhD degree from the National University of Singapore, in 2021. His research interests include the intersection of machine learning and security.



Weiming Zhang received the MS and PhD degrees from the Zhengzhou Information Science and Technology Institute, P.R. China, in 2002 and 2005, respectively. Currently, he is a professor with the School of Information Science and Technology, University of Science and Technology of China. His research interests include information hiding and multimedia security.



Ee-Chien Chang received the PhD degree in computer science from New York University. He is an associate professor with the School of Computing, National University of Singapore (NUS). He was a postdoctoral fellow with DIMACS in Rutgers University and NEC Labs America. His research focuses on cybersecurity security and is particularly intrigued by cross-domain problems. His earlier works include multimedia security, such as image forensic, image watermarking, and biometric cryptography. More recently, he has been investigating issues on security of

machine learning under adversarial environment, and application of hardware-based trusted execution environment.